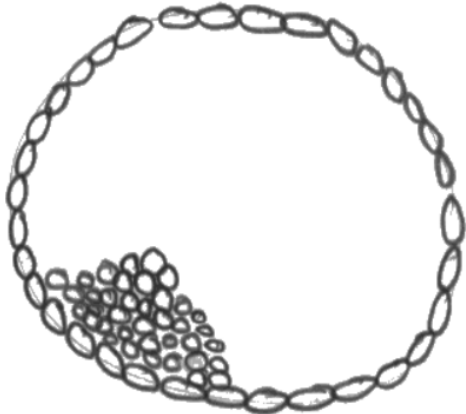


Question Number	Answer	Additional guidance	Mark
6(a)(i)	An answer that makes reference to the following: • (step A) mitosis (1) • (step B) meiosis (1)	accept meiosis I do not accept meiosis I ignore II	(2)

Question Number	Answer	Mark
6(a)(ii)	The only correct answer is A W <i>B is not correct because the acrosome contains digestive enzymes</i> <i>C is not correct because the acrosome contains digestive enzymes</i> <i>D is not correct because the acrosome contains digestive enzymes</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(c)(i)	An answer that includes the following points: • outer layer of cells with internal cell mass (1) • cavity in centre of the diagram (1)	Example of diagram: 	(2)

Question Number	Answer	Mark
6(c)(ii)	The only correct answer is D cells that can give rise to almost any type of cell in the body, excluding totipotent cells <i>A is not correct because pluripotent do not give rise to all cells in the body</i> <i>B is not correct because pluripotent do not give rise to all cells in the body</i> <i>C is not correct because pluripotent do not give rise to totipotent cells</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(d)	An answer that includes the following points: • (stem cells) can {differentiate / specialise} into different {cells / tissues / organs} (1) • great {potential/ importance/ medical implications} of the research in developing medical therapies (1) • salamander embryos do not have a fully developed nervous system (1) • no need to use human embryos (1)	accept named example of a medical therapy use of stem cells in humans e.g. {repair / transplant} {cells / tissues / organs} in humans accept will save human lives / used to treat certain diseases e.g. cancer / heart disease accept salamander embryos do not feel pain / salamanders are not an endangered species / salamanders produce a large number of embryos	(2)